

Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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1. Background & Motivation

Multi-regional clinical trials (MRCTs) are the standard paradigm for global drug development. ICH E17 describes pooling decisions in terms of *similarity of effect modifier distributions* but gives **no quantitative metric, threshold, or procedure**.

Why it matters. With CATE $\tau(x) = E[Y(1) - Y(0) | X = x]$,

$$\hat{\tau}_x = \int \tau(x) dF_x(x),$$

so regions with different effect modifier distributions can observe different treatment effects even when the drug works identically at the individual level.

Gap. The standardized mean difference (SMD) only detects *location* shifts. $N(50, 5^2)$ vs. $N(50, 15^2)$ yields SMD= 0 despite a 3-fold difference in spread; scale and shape differences are invisible.

2. Research Question & Objective

Objective. A distributional dissimilarity index that (i) captures features beyond location, (ii) has a clinically interpretable scale, and (iii) carries a theoretical link to treatment-effect heterogeneity.

Approach. Build the index on Wasserstein-1 distance, normalize by a robust scale, and translate distributional distance to a clinical-scale heterogeneity bound. Validate via simulation; demonstrate via GUSTO-1 IPD.

3. The nABCD Dissimilarity Index

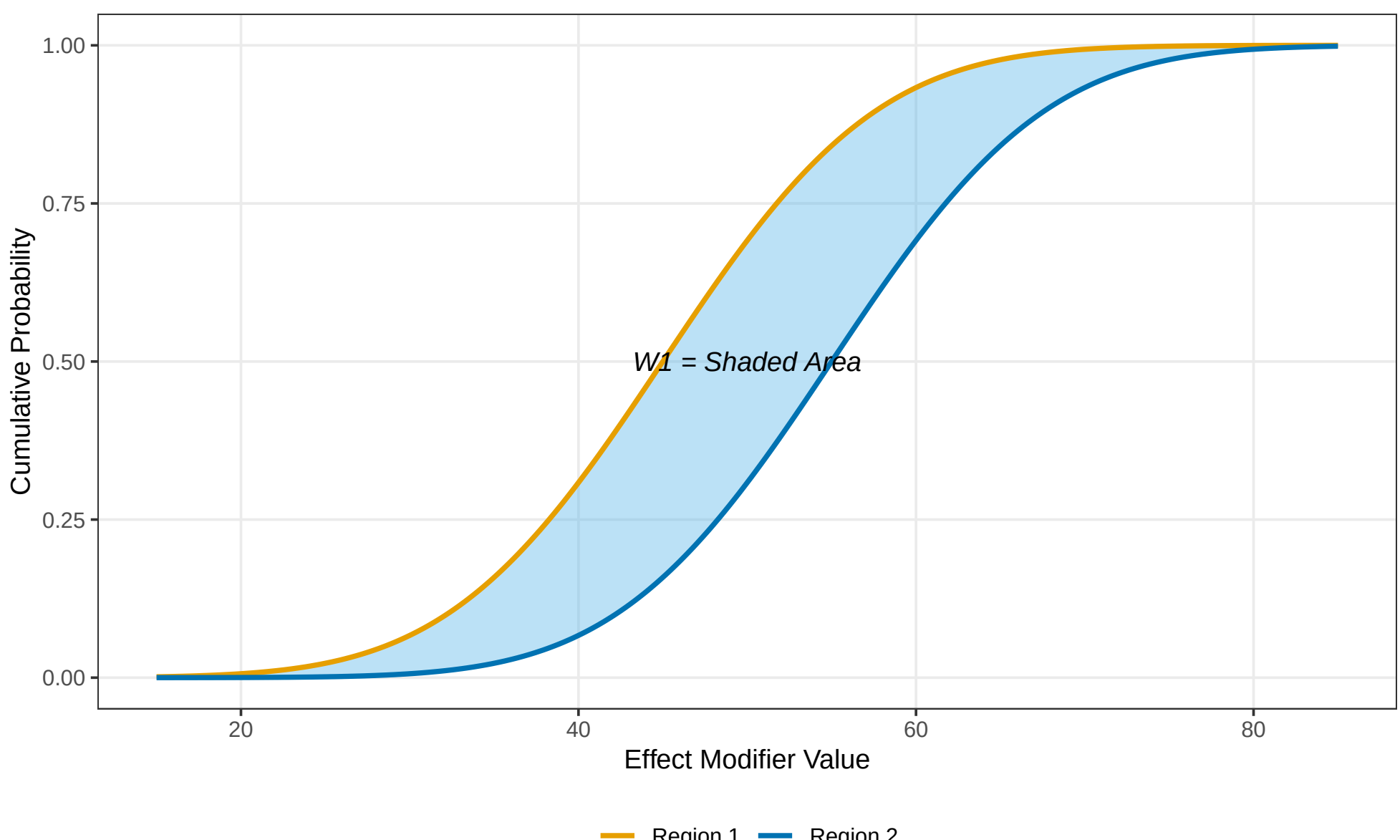
Wasserstein-1 distance between CDFs:

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)| dx,$$

normalized by a robust scale:

$$nABCD(F_1, F_2) = \frac{W_1(F_1, F_2)}{2 \cdot IQR_{pooled}}$$

where IQR_{pooled} is computed from $F_{pooled} = \lambda F_1 + (1 - \lambda) F_2$, $\lambda = n_1 / (n_1 + n_2)$. The factor of 2 calibrates so that a 1-IQR location shift yields $nABCD = 0.5$.



Shaded area between two CDFs = $W_1(F_1, F_2)$.

Properties.

- ≥ 0 ; $= 0 \iff F_1 = F_2$; symmetric; scale-free.
- Responds to location, scale, skewness, and shape.
- Requires only *baseline* effect modifier data – from prior trials, registries, or RWE.

4. Heterogeneity Bound & L^* Calibration

By Kantorovich–Rubinstein duality, if $\tau(x)$ is Lipschitz with constant L ,

$$|\hat{\tau}_1 - \hat{\tau}_2| \leq L \cdot W_1(F_1, F_2),$$

giving the **clinical heterogeneity bound**

$$\mathbf{L}_{max} = 2 \cdot L \cdot IQR_{pooled} \cdot nABCD(F_1, F_2)$$

with bootstrap CI by linear mapping. L is the per-unit CATE slope from Phase 2 interaction analysis or subgroup meta-analyses.

When L is unknown, rearrange to obtain the **required CATE sensitivity**:

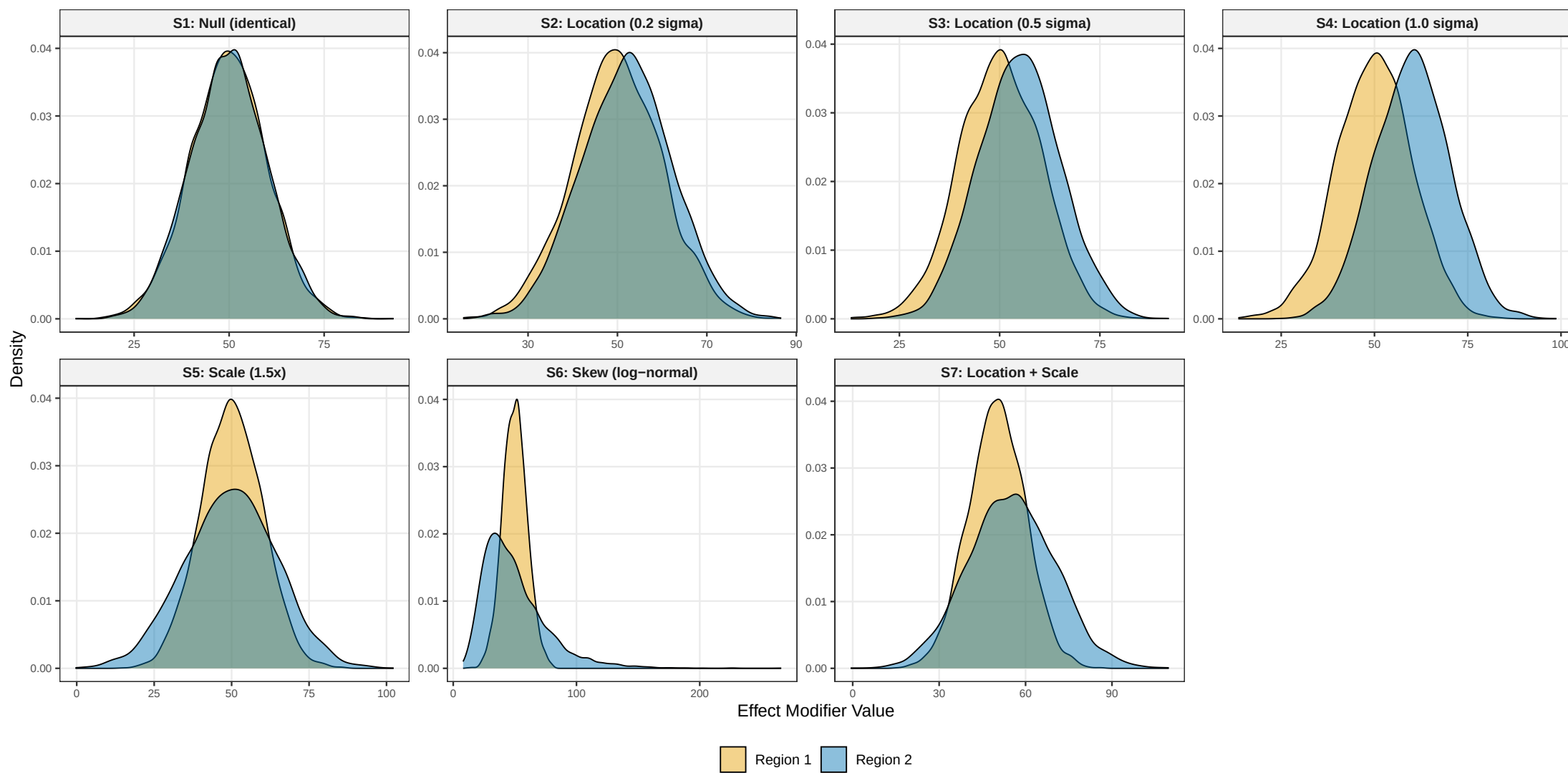
$$L^* = \frac{\mathbf{L}_{min}}{2 \cdot IQR_{pooled} \cdot nABCD}$$

L^* is the slope at which the observed distributional difference would produce a clinically meaningful $\mathbf{L}_{min}$. The sponsor judges plausibility of L^* for the effect modifier and drug class.

A W_1 -specific property: W_2 and KS admit no analogous bound (Gibbs & Su, 2002). **Inference.** Percentile bootstrap ($B \geq 2,000$); BCa rejected for boundary behavior. Recommended $n \geq 100$ per region.

5. Simulation Design (7 Scenarios)

Systematic scenarios isolating location, scale, and skew. 10,000 replications each; $n \in \{50, 100, 200\}$ per region; $B = 2,000$ bootstrap resamples.



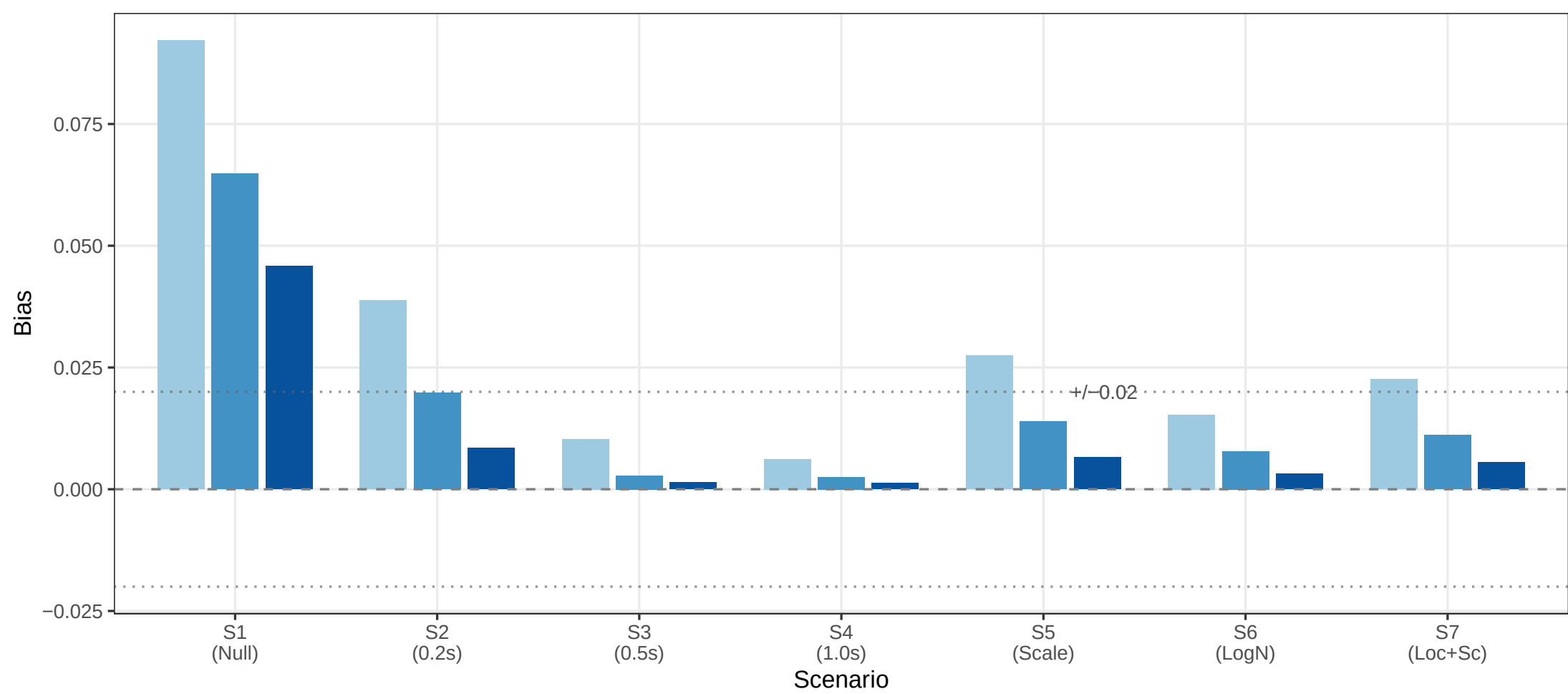
Scenario overview: location, scale, and skew alterations relative to the null reference.

ID	Type	F_1 vs. F_2	True
S1	Null	$N(50, 10^2)$ vs. same	0.000
S2	Loc. 0.2x	vs. $N(52, 10^2)$	0.073
S3	Loc. 0.5x	vs. $N(55, 10^2)$	0.180
S4	Loc. 1.0x	vs. $N(60, 10^2)$	0.328
S5	Scale 1.5x	vs. $N(50, 15^2)$	0.122
S6	Skew (LogN)	vs. $1.5xN(\mu, \sigma^2)$	0.304
S7	Loc. + Scale	vs. $N(55, 15^2)$	0.175

S7 is the most realistic MRCT pattern; S6 matches biomarker CVs= 53%.

6. Estimation Quality: Bias & Coverage

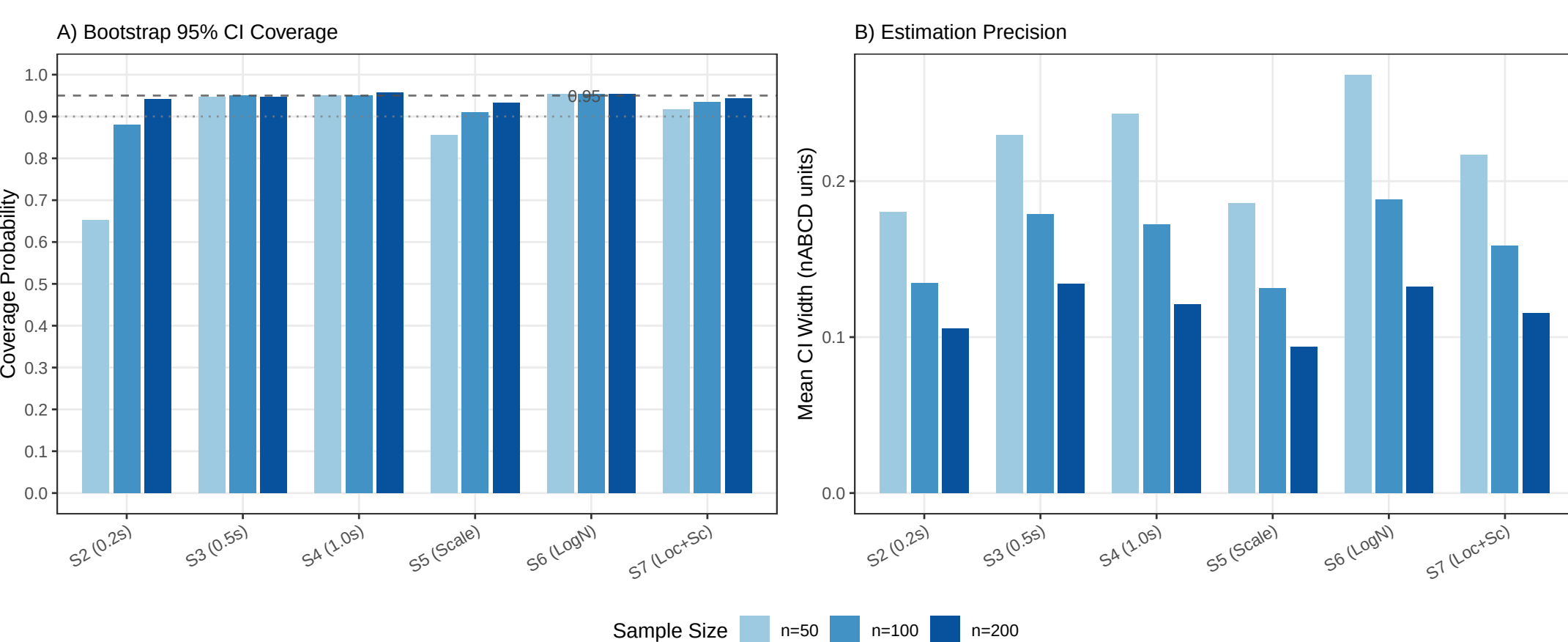
Bias and 95% percentile bootstrap CI coverage at $n = 100$ per region:



Sample Size: $n=50$, $n=100$, $n=200$

Bias of $nABCD$ across scenarios and sample sizes.

Scenario	True	Bias	Coverage
S1 (Null)	0.000	0.066	~1
S2 (0.2x)	0.073	0.020	0.981
S3 (0.5x)	0.180	0.003	0.951
S4 (1.0x)	0.328	0.003	0.950
S5 (Scale)	0.122	0.014	0.910
S6 (Skew)	0.304	0.008	0.954
S7 (Loc+Scale)	0.175	0.011	0.935



Coverage and precision summary across scenarios.

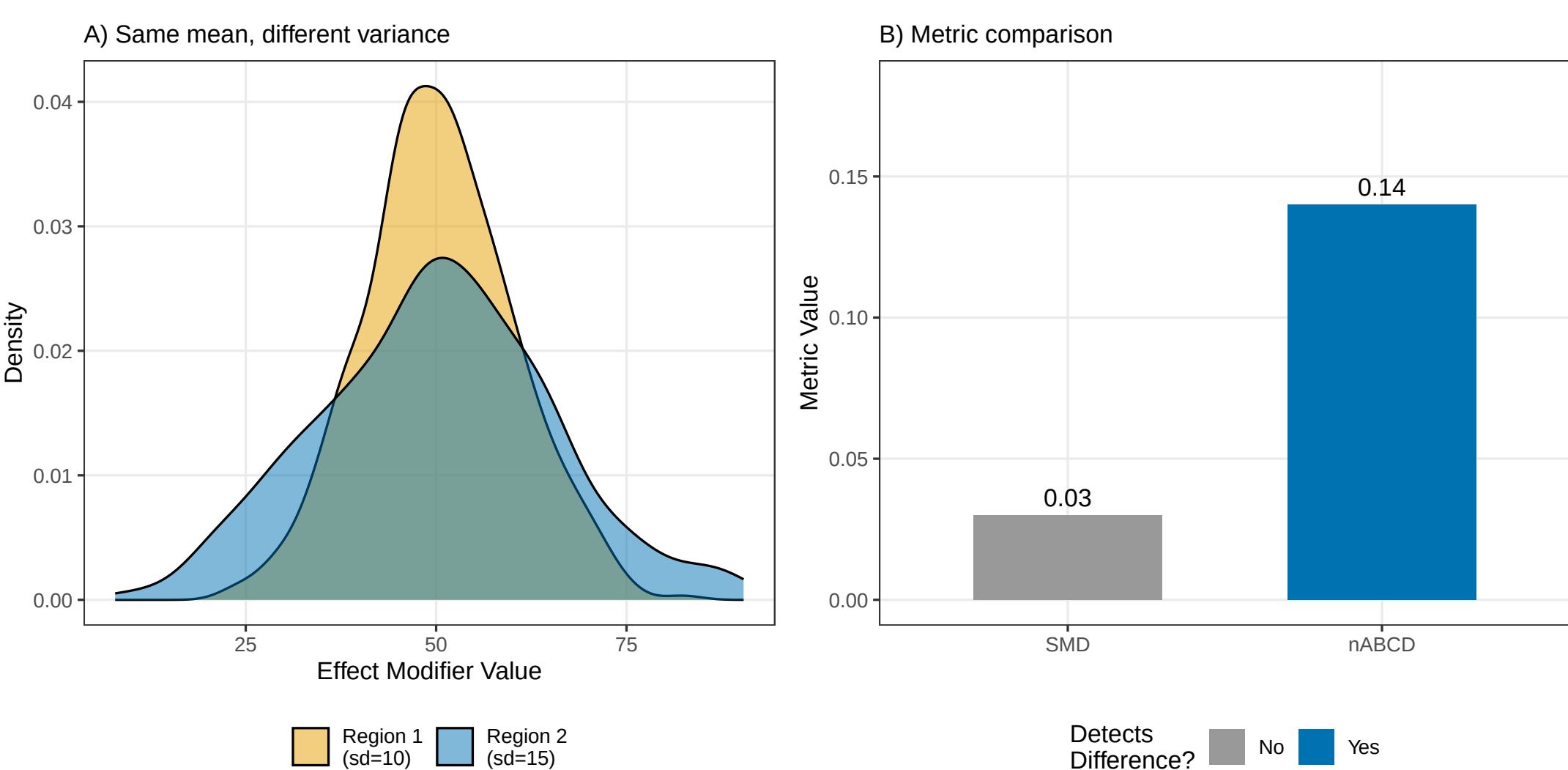
¹Near boundary true value; non-negativity constraint induces positive bias (Andrews, 2000).

Take-aways. True $nABCD \geq 0.1$: bias < 0.02, coverage 0.88–0.96 at $n \geq 100$. **Percentile bootstrap** preferred over BCa.

7. nABCD vs. SMD: Beyond Location

At $n = 100$ per region, SMD is blind to precisely the features that drive non-linear effect-modifier heterogeneity:

Scenario	nABCD	SMD	Detected by
S3 Location	0.183 ± 0.048	0.50 ± 0.14	Both
S5 Scale	0.136 ± 0.033	0.00 ± 0.14	Only nABCD
S6 Skew	0.312 ± 0.048	0.00 ± 0.14	Only nABCD



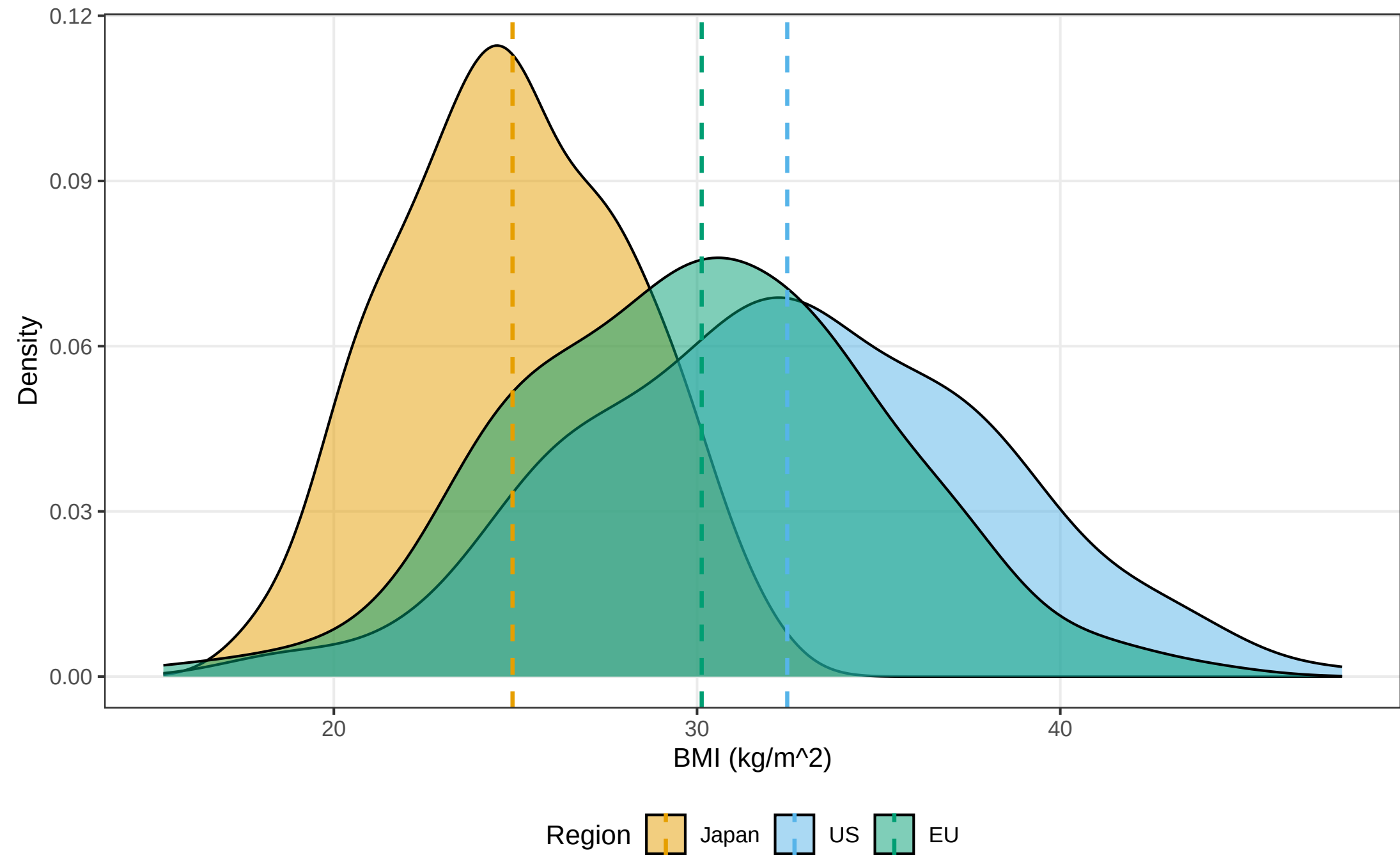
nABCD detects scale and skew where SMD stays flat.

Framework philosophy. Estimation-centered, not testing. Report $nABCD$, bootstrap CI, and $\mathbf{L}_{max}$ or L^* – never a binary verdict. No universal cutoff: clinical weight depends on L , external to the distributional comparison. ICH E17 aligned.

8. GUSTO-1 Application: Drug T MRCT Planning

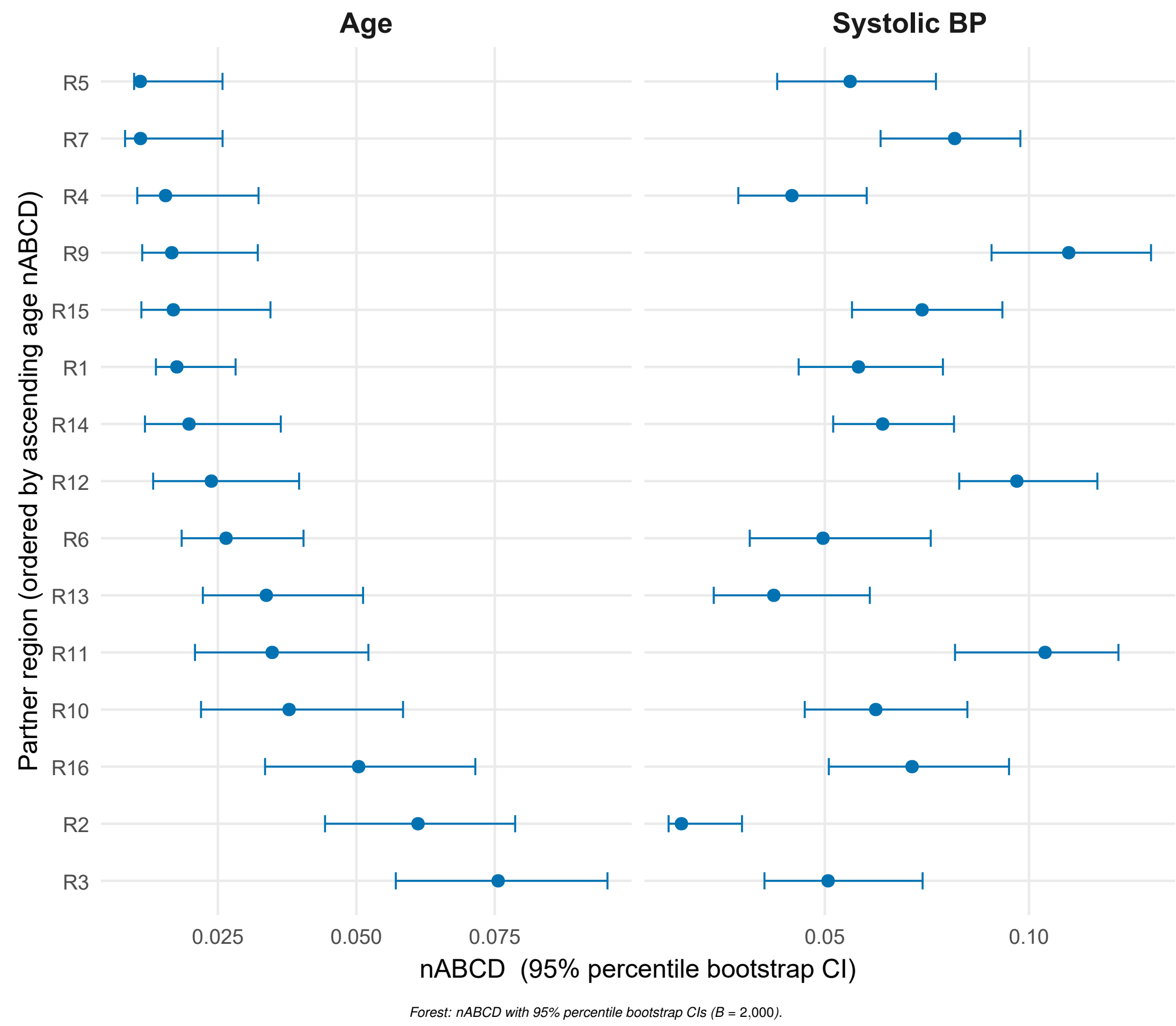
A sponsor develops **Drug T**, a novel thrombolytic for acute myocardial infarction, and plans a Phase 3 MRCT. At the planning stage the sponsor uses IPD from **GUSTO-1** ($N = 40,830$; 16 anonymized regions) to assess distributional similarity of candidate effect modifiers across regions and to identify pooling partners for a designated anchor region.

Candidate effect modifiers. Age and systolic blood pressure (SBP), flagged by Drug T's Phase 2 exploratory analyses; class-level numerical CATE slopes are unavailable. L is therefore treated as unknown and the L^* pathway applies. **Region 8** is the anchor ($n = 2,916$).

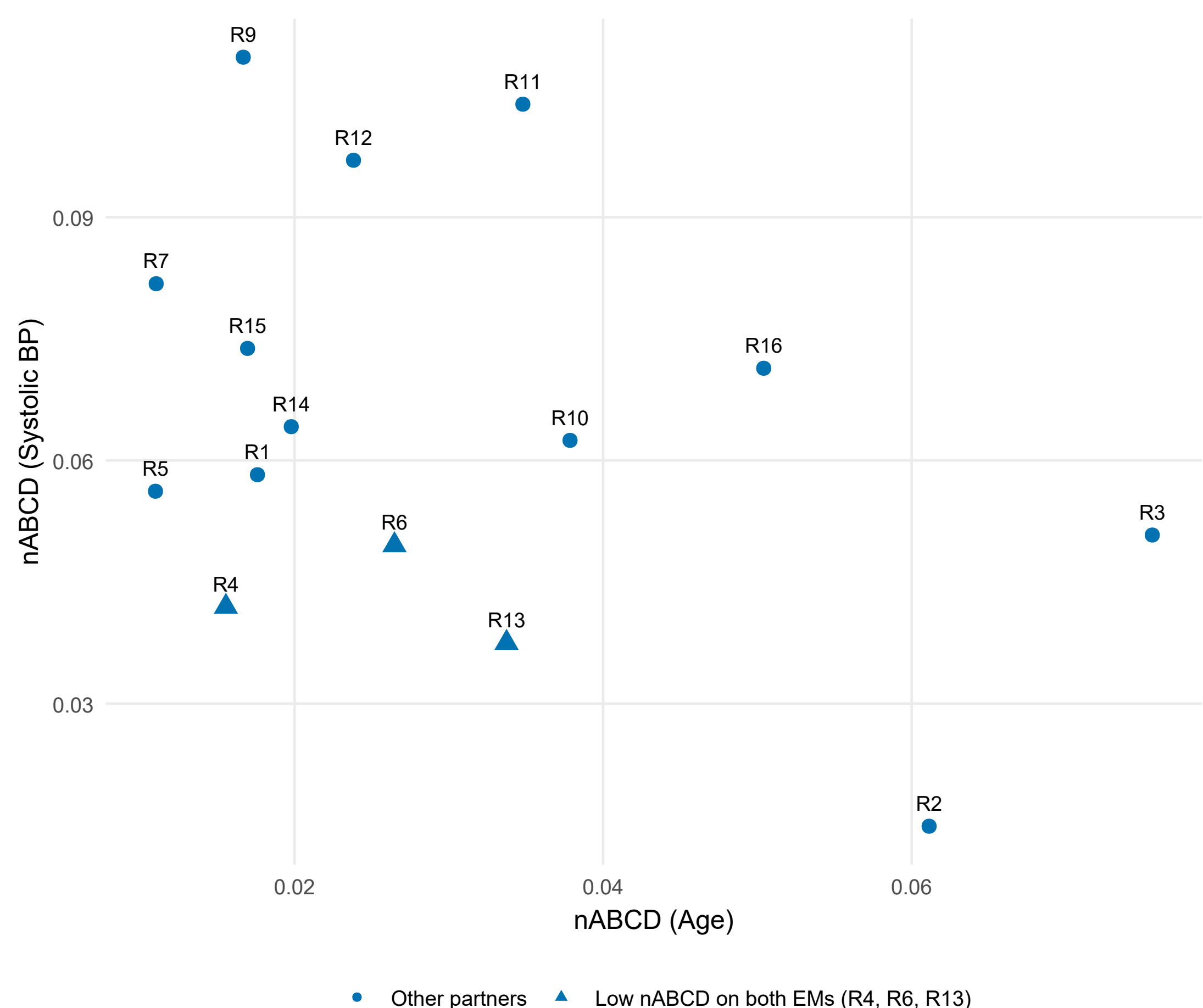


GUSTO-1 anchor (R8) vs. partner regions: distributional overview.

Rank	Partner	n	nABCD _{age}	nABCD _{sbp}
1	R6	1909	0.011 [0.010, 0.026]	0.056 [0.038, 0.077]
2	R7	3150	0.011 [0.008, 0.026]	0.082 [0.064, 0.098]
3	R4	2876	0.016 [0.010, 0.032]	0.042 [0.029, 0.060]
4	R9	3123	0.017 [0.011, 0.032]	0.110 [0.091, 0.130]
14	R2	2952	0.061 [0.044, 0.079]	0.015 [0.012, 0.030]
15	R3	2030	0.076 [0.057, 0.095]	0.051 [0.035, 0.074]



Forest: nABCD with 95% percentile bootstrap CIs ($B = 2,000$).



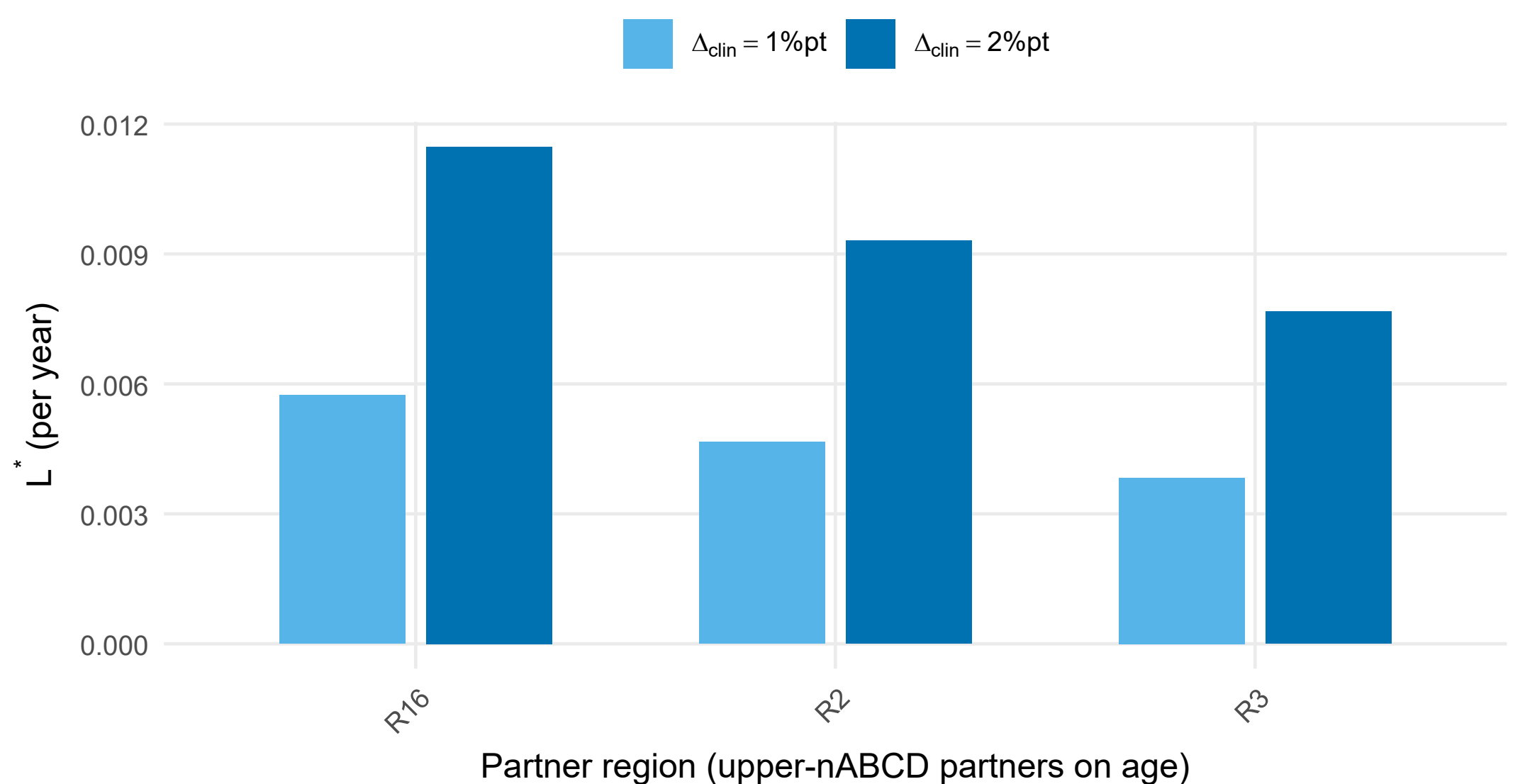
Other partners (blue dots), Low nABCD on both EMs (R4, R6, R13) (blue triangles).

Joint view: lower-left favors pooling on both EMs.

R2 vs. R9 asymmetry. R2 has the smallest SBP nABCD (0.015) but the 2nd-largest age nABCD (0.061); R9 shows the opposite. A single-modifier ranking would draw opposite conclusions.

Partner	nABCD _{age}	nABCD _{sbp}	L^* age	L^* SBP
R4	0.016	0.042	0.0180	0.0038
R6	0.008	0.050	0.0110	0.0030
R13	0.004	0.037	0.0087	0.0042
R9	0.017	0.110	0.0171	0.0015
R2	0.061	0.015	0.0047	0.0105

(a) Age: Sensitivity analysis



(b) Systolic BP: Sensitivity analysis

L^* at 1%pt (light blue), L^* at 2%pt (dark blue)